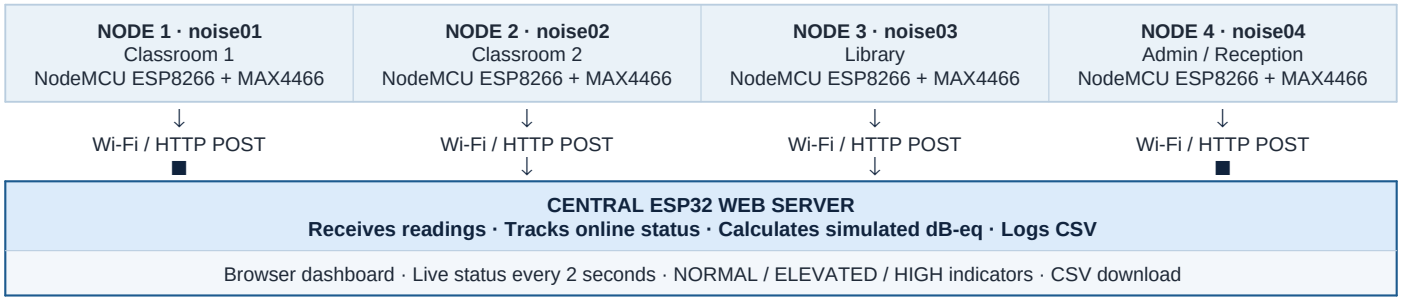
SCHOOL NOISE MONITORING SYSTEM

Live multi-location sound-indicator prototype

PROJECT OVERVIEW

Four NodeMCU ESP8266 stations with MAX4466 microphone modules sample ambient sound and send readings over school Wi-Fi to a central ESP32. The ESP32 hosts a live dashboard and stores received readings in a downloadable CSV log.

SYSTEM STRUCTURE



HOW THE SYSTEM WORKS

1. Each microphone station samples its analog output and calculates an AC-RMS indicator.
2. The NodeMCU sends its station ID, place, RMS and peak readings to the ESP32 over HTTP.
3. The ESP32 updates the relevant station status and appends readings to its local CSV file.
4. A browser on the same reachable Wi-Fi/LAN displays the live dashboard.

WHAT VISITORS CAN SEE

- Four station identities and online/offline status.
- Simulated dB-equivalent indicator (dB-eq), raw RMS and signal peak.
- Provisional green/orange/red categories and last-update time.
- Downloadable CSV log of readings received by the central ESP32.

NOTE: The displayed dB-eq value is a simulated educational indicator calculated from microphone RMS ADC counts using an arbitrary reference. It is NOT calibrated dB(A), does not represent a verified sound-pressure level, and must not be used to claim regulatory compliance. Actual dB(A) reporting requires calibration and validation against a reference sound-level meter. Colour categories are demonstration indicators, not official noise-limit determinations.

PROJECT AIM: Demonstrate multi-location sensing, wireless data transmission, live visualization, and data logging as a foundation for future calibrated environmental monitoring.